

Group - A

* Aim:- To determine the viscosity of given sample of lubricating oil with Redwood viscometer No.1 and to study the viscosity at various temperatures.

* Articles Required:- Redwood viscometer No.1, Thermometers, Oil Sample, 50ml Measuring cylinder etc.

* Significance of determination:- Viscosity is the property of a homogeneous liquid which causes it to offer frictional resistance to its motion. Viscosity of lubricating oil tells about its suitability for lubricating purpose.

The function of a lubricant is to reduce the friction between the moving parts & to avoid the direct metal to metal contact.

The moving parts remain apart due to extremely thin liquid film in between. The frictional resistance therefore, is entirely on account of the shearing of liquid which is in between the metallic surfaces. The viscosity of fluid measures the amount of internal resistance of lubricating oil.

The object of this test is to ascertain, whether the lubricating oil is sufficiently viscous to adhere to the bearings to which it is applied or is so thin that it may be squeezed out of the bearing due to high pressure & temperature.

* Theory:- Viscosity is that property of a fluid which is due to the resistance to its flow. It is expressed in 3 ways:-

- (a) Absolute dynamic viscosity expressed in C.G.S. unit as poise.
 (b) Absolute kinematic viscosity expressed in C.G.S. unit as stoke.
 (c) Time taken by the determining instrument in seconds noted by the name of Instrument (Redwood viscosity).

a.) Absolute dynamic viscosity :- It is the tangential force on unit area of either of two parallel layers at unit distance apart when the space between the planes is filled with the fluid & one of the planes moves relatively to the other with unit velocity in its own plane. It is defined as the ratio of the shearing stress to the rate of shearing strain. If force F is required to keep moving an article of surface area A , in contact with fluid separated from a stationary surface by thickness ' d ' with velocity ' v ', then,

$$\text{Shearing stress} = F/A.$$

$$\text{Rate of shearing strain} = v/d$$

$$\text{Absolute viscosity} = F \cdot d / A \cdot v$$

When η is in poise, then $F = 1 \text{ dyne}$, $A = 1 \text{ sq. cm}$, $d = 1 \text{ cm}$,
 $v = 1 \text{ cm per second}$

b.) Absolute kinematic viscosity :- It is obtained by dividing Dynamic viscosity by density. The metric units are stoke & centistokes. Thus, if ' ρ ' is density, then
 Kinematic viscosity = $\frac{\eta}{\rho}$ Stoke.

c.) Redwood viscosity:- The timing or the rate of flow of given quantity of liquid from a given height through a calibrated capillary tube under its own weight may be used as a means for measuring viscosity & called after the name of the apparatus used for its determination. The time in seconds required for 50 mL of oil to gravitate through the Redwood viscometer at a given temperature is expressed as its viscosity in Redwood seconds at that temperature.

Absolute dynamic & kinematic viscosities can also be determined from the Redwood values by means of the following formula:-

$$\bullet \text{ Absolute dynamic viscosity} = \frac{(At - B) \times \rho}{t}$$

$$\bullet \text{ Kinematic viscosity} = \frac{\mu}{\rho} = \frac{At - B}{t}$$

When A and B are constants known as instrument factors, 't' is Redwood viscosity & 'ρ' is the density of fluid. The values of A & B for Redwood viscometer No. 1 are as follows:-

	A	B
Between 34-100 seconds	1.79	0.00260
Above 100 seconds	0.50	0.00247

Water bath must be filled with water before starting the experiment.

Difference between Redwood No. 1 and No. 2:-

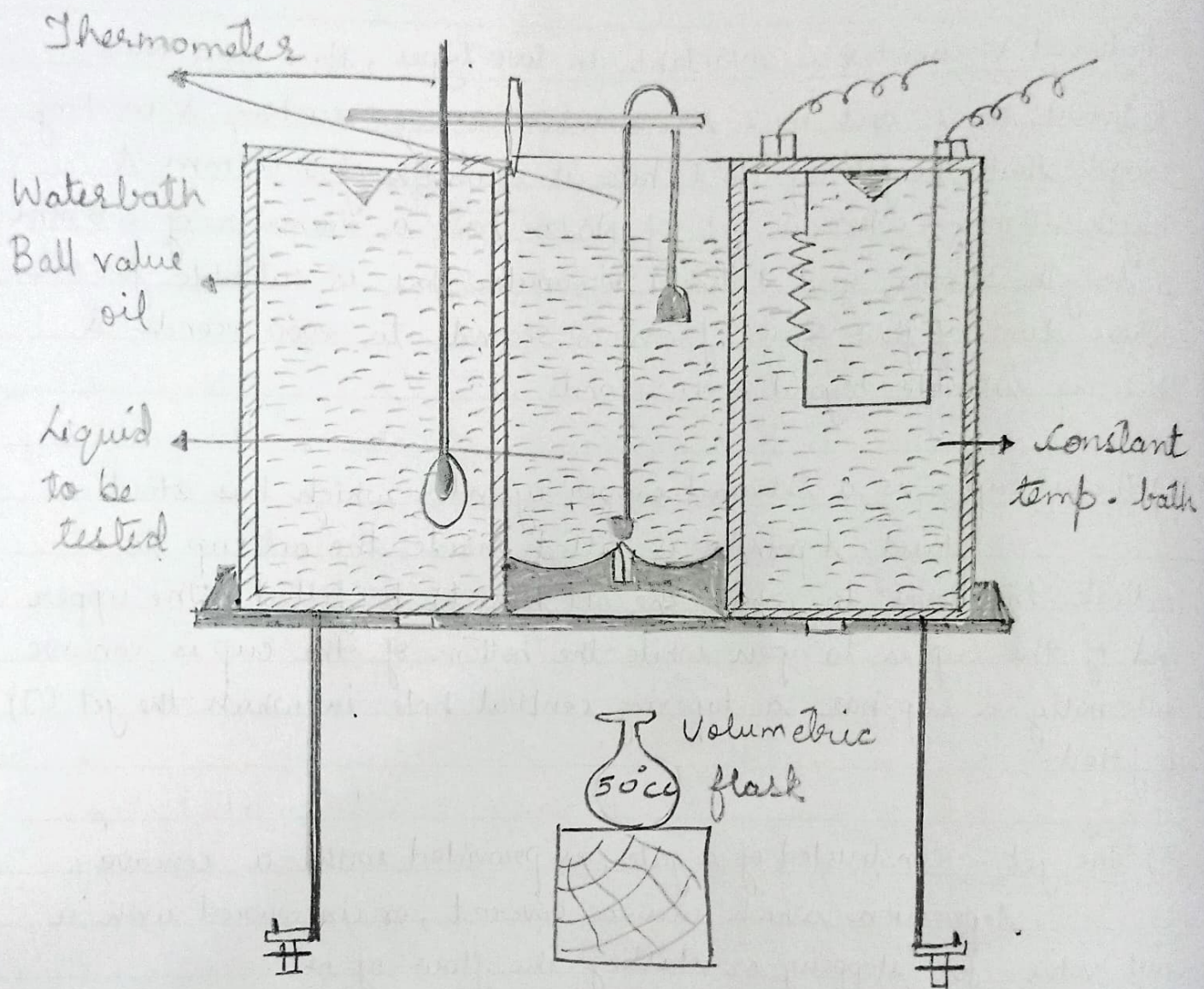
→ Redwood viscometer is available in two types, No. 1 and No. 2

Redwood No. 1 and No. 2 are similar in construction & working except that the jet of No. 1 has a diameter of 1.62 mm & length 10 mm. Whereas jet of No. 2 has a diameter of 3.8 mm & length 50 mm and Redwood viscometer No. 1 is suitable for oils whose time of flow varies from 30 seconds to 2000 seconds & No. 2 is suitable beyond 2000 seconds.

1.) The oil cup:- is a silvered copper cylinder which has stout stand. A pointer is fitted inside the oil cup to indicate the level to which the oil has to be filled. The upper end of the cup is to open while the bottom of the cup is concave internally & cup has a tapering central hole in which the jet (J) is fitted.

2.) The jet:- constructed of a gate, is provided with a concave depression which can be covered, or uncovered with a ball valve for stopping or starting the flow of oil.

3.) Water bath:- is made up of brass & surrounds the oil cup. It is heated electrically, is provided with a tap for emptying water. Stirring of the bath is effected by means of cylindrical surrounding the oil cup & is provided with 4 vanes.



Redwood Viscometer No. 1

4.) Thermometers -

- Two thermometers to read the temperature of oil & water are provided.
- The apparatus rests on a tripod stand sufficient space for receiving 50 ml. flanged volumetric flask beneath the jet.

* Procedure :-

- 1) Viscometer is cleaned & leveled using circular spirit level and base screws the tripod.
- 2) The ball valve is fixed in position in the oil cup & the oil cup filled up to the level mark with oil.
- 3) The water bath is then filled with water nearly up to the point of oil level in the cup.
- 4) The thermometers are inserted in place provided in water bath & oil cup.
- 5) Now, carefully place the 50 ml measuring cylinder exactly in the centre below the jet.
- 6) Keep the oil & water well stirred & note their temperature.
- 7) When the temperature of oil & water becomes same & steady at room temperature, raise the ball valve & suspend it from the thermometer bracket & allow the oil to flow into the measuring cylinder kept below.
- 8) Simultaneously, start a stop watch or look at your wrist watch when the level of oil flowing into the measuring cylinder reaches 50 ml. mark, note the time in seconds elapsed to collect 50 ml of oil.

Observation :-

S.NO.	Temperature in $^{\circ}\text{C}$		Time in seconds for flow of 50ml oil
	Water	Oil	
1.	37 $^{\circ}$	37 $^{\circ}$	50
2.	45 $^{\circ}$	45 $^{\circ}$	40
3.	54	54 $^{\circ}$	30

- 9.) Replace the ball valve in position to seal the jet to prevent overflow of the oil.
- 10.) Transfer the oil into oil cup from measuring cylinder up to the indicator tip of the cup.
- 11.) Replace properly clean standard measuring cylinder centrally below the jet.
- 12.) The water bath is then heated which in turn heats the oil.
- 13.) When the oil temperature reaches near to the test temperature, stop heating & bring down temperature of water equal to the temperature of oil by running out some hot water through the outlet & adding cold water from the top with continuous stirring.
- 14.) When temperature of both oil as well as water become steady, again collect 50ml of oil and note the time in seconds required.
- 15.) Now repeat the experiment at 5 elevated temperatures. The temperature difference between the two observations should not be less than 5°C & not more than 10°C .
- 16.) The temperature of oil & water should always be same while recording the observation.

* Calculation :-

$$\text{Absolute Kinematic viscosity} = \frac{(A\tau - B) \times \rho}{\tau}$$

where A & B are constants known as instrument factors,
 τ is Redwood viscosity & ρ is the density of fluid

$$A = 1.79 \quad B = 0.00260 \quad \text{for } 34.100 \text{ sec}$$

$$1.) \frac{(1.79)(67)}{67} - 0.0026 = 1.79 \quad (t = 67)$$

$$2.) \frac{(1.79)(46)}{46} - 0.0026 = 1.789 \quad (t = 46)$$

$$3.) \frac{(1.79)(42)}{42} - 0.0026 = 1.789 \quad (t = 42)$$

* Result:- Redwood viscosity of the given oil at temperature is
1.789 Stokes

* Precautions:-

- 1.) Time should be noted carefully for collecting 50 ml of oil.
- 2.) Temperature should be steady & constant while taking observations.
- 3.) Difference in the temperature of water bath & oil should not be more than 2°C while noting time.

* Aim:- To determine the viscosity of given sample of lubricating oil with Redwood viscometer No.2 & to study the viscosity at various temperatures.

* Articles Required:- Redwood viscometer, thermometers, oil sample, measuring flask 50ml.

* Significance of determination:- Viscosity is the property of a homogeneous liquid which causes it to offer frictional resistance to its motion. Viscosity of lubricating oil tells about its suitability for lubricating purpose.

The function of a lubricant is to reduce the friction between the moving parts & to avoid the direct metal to metal contact.

The moving parts remain apart due to extremely thin liquid film in between. The frictional resistance therefore is entirely on account of the shearing of liquid which is in between the metallic surfaces. The viscosity of fluid measures the amount of this internal resistance of lubricating oil.

The object of this test is to ascertain, whether the lubricating oil is sufficiently viscous to adhere to the bearings to which it is applied or is so thin that it may be squeezed out of the bearing due to high pressure & temperature.

* Theory/Principle:- Viscosity is the property of a fluid which is due to the resistance to its flow. It is expressed in 3 ways:-

- (a) Absolute dynamic viscosity expressed in C.G.S unit as poise.
 (b) Absolute kinematic viscosity expressed in C.G.S unit as stoke
 (c) Time taken by the determining instrument in seconds noted by the name of instrument. (Redwood viscosity).

a.) Absolute dynamic viscosity :- It is the tangential force on unit area of either of two parallel layers at unit distance apart when the space between the planes is filled with the fluid & one of the planes moves relatively to the other with unit velocity in its own plane.

It is defined as the ratio of the shearing stress to the rate of shearing strain. If F is force required to keep moving an article of surface area A in contact with the fluid separated from a stationary surface by thickness d with velocity v then,

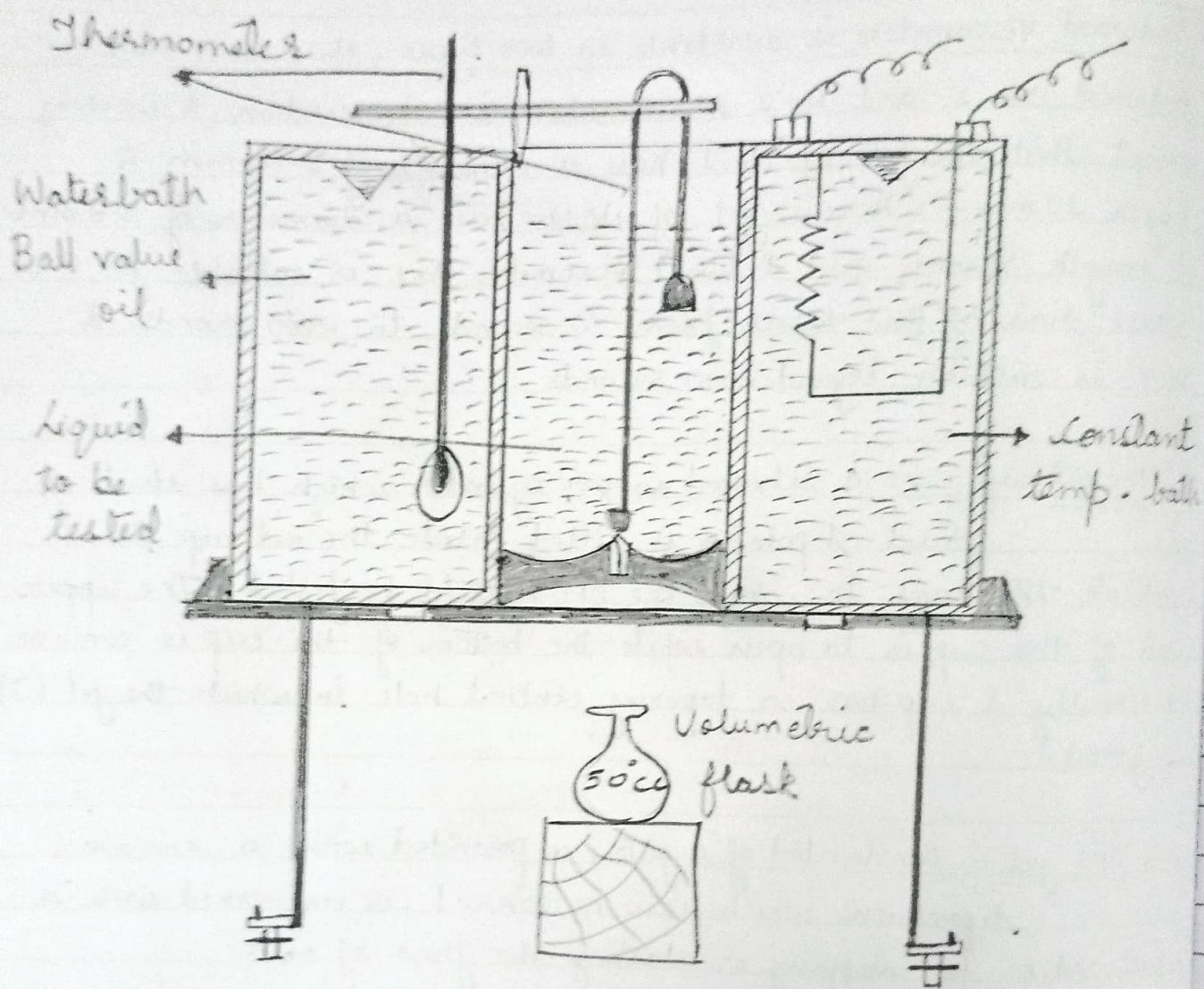
$$\text{Shearing stress} = F/A$$

$$\text{Rate of shearing strain} = v/d$$

$$\text{Absolute viscosity} = Fd/A \cdot v$$

When η is in poise, then $F = 1 \text{ dyne}$, $A = 1 \text{ sq. cm}$, $d = 1 \text{ cm}$,
 $v = 1 \text{ cm per second}$.

b.) Absolute kinematic viscosity :- It is obtained by dividing Dynamic viscosity by density. The metric units are stoke & centistokes. Thus, if ρ is density, then kinematic viscosity = $\frac{\eta}{\rho}$ stoke.



Redwood Viscometer NO. 2

C.) Redwood viscosity :- The timing or the rate of flow of given quantity of liquid from a given height through a calibrated capillary tube under its own weight may be used as a means for measuring viscosity & called after the name of the apparatus used for its determination.

The time in seconds required for 50 ml of oil to gravitate through the jet Redwood viscometer at a given temperature is expressed as its viscosity in Redwood seconds at that temperature.

Absolute dynamic & kinematic viscosities can also be determined from the Redwood values by means of the following formula:-

$$\text{Absolute dynamic viscosity } \mu = \frac{(At - B) \times \rho}{t}$$

$$\text{Kinematic viscosity} = \frac{\mu}{\rho} = \frac{At - B}{t}$$

Where A & B are constants known as instrument factors, t is Redwood viscosity & ρ is the density of fluid. The values of A & B for Redwood Viscometer No. 1 are as follows:-

	A	B
Between 34-100 seconds	1.79	0.00260
Above 100 seconds	0.50	0.00247

Water bath must be filled with water before starting the experiment.

Difference between Redwood No.1 and No.2:

Redwood viscometer is available in two types No.1 and No.2. Redwood No.1 and No.2 are similar in construction and working except that the jet of No.1 has a diameter of 1.62 mm & length 10 mm whereas jet of No.2 has a diameter of 3.8 mm & length 50 mm and Redwood viscometer No.1 is suitable for oils whose time of flow varies from 30 seconds to 2000 seconds & No.2 is suitable beyond 2000 seconds.

The apparatus consists of following parts:-

- 1.) The oil cup:- is a silvered copper cylinder which has stout stand. A pointer is fitted inside the oil cup to indicate the level to which the oil is to be filled. The upper end of the cup is open while the bottom of the cup is concave internally & cup has a tapering central hole in which the jet (J) is fitted.
- 2.) The jet:- constructed of a gate is provided with a concave depression which can be covered, or uncovered with a ball valve for stopping or starting the flow of oil.
- 3.) Water bath:- is made of brass & surrounds the oil cup. It is heated electrically. It is provided with a tap for emptying water. Stirring of the bath is affected by means of cylinder surrounding the oil cup & is provided with 9 vanes.

- 8.) Replace the ball valve in position to seal the jet to prevent overflow of the oil.
- 9.) Transfer the oil into oil cup from measuring cylinder up to the indicator tip of the cup.
- 10.) Properly replace clean standard measuring cylinder centrally below the jet.
- 11.) The water bath is then heated which in turn heats the oil.
- 12.) When the temperature of oil reaches near to the test temperature stop heating and bring down the temperature of water equal to it by running out some hot water through the outlet & adding cold-water from the top while stirring continuously.
- 13.) When temperature of both oil as well as water become steady & constant, again collect 50 ml. of oil & note the time in seconds required to collect 50 ml of oil.
- 14.) Now, repeat the experiment at 5 elevated temperatures. The temperature difference between the two observations should not be less than 5°C & more than 10°C .
- 15.) The temperature of oil & water should always be same, steady & constant while recording each observation.

* Calculation :-

$$\text{Absolute kinematic viscosity} = \frac{(A - B) \times p}{t}$$

Where A & B are constants known as instrument factors of Redwood viscosity & p is the density of fluid.

Observation :-

S.NO.	Temperature in $^{\circ}\text{C}$		Time in seconds for flow of 50ml oil
	Water	Oil	
1.	37°	37°	50
2.	45°	45°	40
3.	54°	54°	30

- 8.) Replace the ball valve in position to seal the jet to prevent overflow of the oil.
- 9.) Transfer the oil into oil cup from measuring cylinder up to the indicator tip of the cup.
- 10.) Properly replace clean standard measuring cylinder centrally below the jet.
- 11.) The water bath is then heated which in turn heats the oil.
- 12.) When the temperature of oil reaches near to the test temperature stop heating and bring down the temperature of water equal to it by running out some hot water through the outlet & adding cold-water from the top while stirring continuously.
- 13.) When temperature of both oil as well as water become steady & constant, again collect 50 ml. of oil & note the time in seconds required to collect 50 ml of oil.
- 14.) Now, repeat the experiment at 5 elevated temperatures. The temperature difference between the two observations should not be less than 5°C & more than 10°C .
- 15.) The temperature of oil & water should always be same, steady & constant while recording each observation.

* Calculation -

$$\text{Absolute kinematic viscosity} = \frac{(A - B) \times p}{t}$$

Where A & B are constants known as instrument factors of Redwood viscosity & p is the density of fluid.

$$\begin{aligned}\text{Absolute Kinematic viscosity} &= \frac{1.79 \times 40}{40} - 0.0026 \\ &= 1.7899 \text{ Stokes.}\end{aligned}$$

★ Result :- Redwood viscosity of the given oil is 1.789 stokes.

★ Precautions :-

- 1.) Time should be noted carefully everytime during collection of 50 ml of oil.
- 2.) Temperature should be steady while noting time for flow.
- 3.) Difference in the temperature of water bath & oil should not vary even by 1°C while noting time.

* Aim:- To determine flash point of illuminating or lubricating oil by Abel's apparatus.

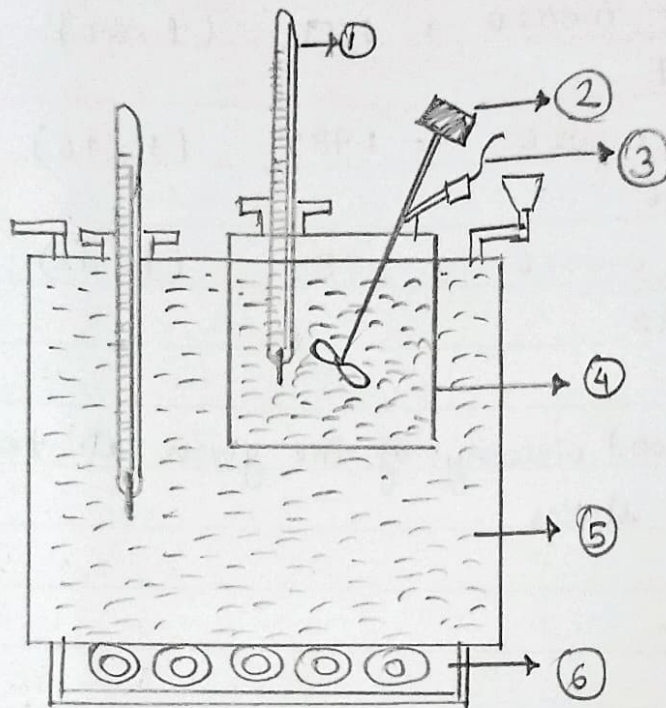
* Articles required:- Abel's apparatus, thermometers, oil sample, water & LPG etc.

* Definition:- Flash point of oil is defined as the minimum temperature at which it gives off sufficient vapours, which when mixed with air forms an ignitable mixture & gives a momentary flash of light on the application of a small flame.

* Significance of determination:- The determination of flash point is of great importance in case of lubricating & illuminating oils. If the oils are sufficiently volatile at ordinary temperatures, the issuing vapours will form an explosive mixture with air & cause fire hazards. To ensure safety while using oils, certain minimum temperatures are laid down for illuminating & lubricating purposes, below which it should not give off inflammable vapours.

The flash point of oil shows the extent of more volatile constituents in the oil. The lower the flash point, the greater is the amount of more volatile constituents.

The flash point is clearly distinguished from fire point, which is the temperature at which the air space above the oil has sufficient concentration of vapours so that it burns continuously for about five seconds when brought in contact with the flames.



1. Thermometer
2. Stirrer
3. Flame point
4. Oil cup
5. Water
6. Heater

* Parts of Apparatus:- The Abel's apparatus is a cylindrical structure consisting of the following parts:-

- 1) The oil cup:- This is a cylindrical vessel, made of brass, open at the top end fitted on the outside with a flat circular flange. Projecting at right angles, within the cup there is a gauge consisting of a piece of wire bent upwards & terminating in a point. The oil is filled up to this point.
- 2) Cover:- This is fitted with a stirrer & thermometer socket made of brass mounted on the oil cup.
- 3) Heating vessel:- The heating vessel consists of two flat bottomed cylindrical vessels, placed coaxially one inside the other & soldered at their tops to a flat copper ring. The space between the two vessels thus totally enclosed & is used as a water jacket.

In addition to above, the working with Abel's apparatus requires the heating arrangement & thermometers of appropriate range. It may be heated electrically or with a burner.

- * Procedure:-
- 1) Fill the oil cup with the oil under test up to the indicator mark. Replace the cover.
 - 2) Fix the oil cup into the apparatus & assemble the paddle stirrer & thermometers with its bulb dipping into the oil & water at their respective places, provided for in the apparatus.
 - 3) Fill the water bath with cold water.

Observation - 01

S.NO.	Temperature	Observation
1.	Room Temperature (36°)	NO flash
2.	37	No flash
3.	39	NO flash
4.	41	No flash.
5.	43	Flash

Observation - 02

S.NO.	Temperature	Observation
1.	40	NO flash
2.	41	No flash
3.	42	No flash.

- 4.) Close the sliding shutter & light the standard flame & start heating the water bath with the help of burner.
- 5.) Stir the oil continuously by turning the paddle stirrer. Stirring should be discontinued only during the introduction of the test flame over the oil surface.
- 6.) At every degree, rise of oil temperature open the sliding shutter & introduce the test flame over the oil surface through the central opening to see whether the oil gives a flash or not.
- 7.) Record the minimum temperature at which a distinct flash appears as the flash point of the given oil.
- 8.) Now, cool the oil by 5°C & verify the flash point again by every 1°C rise of temperature.

* Result:- The flash point of given oil sample is 43°C

* Precautions:-

- 1.) The bluish flame surrounding the test flame should not be confused with the true flame
- 2.) The traces of solvent used for cleaning the apparatus should be removed completely.